

Bandgap Reference (BGR) – SKY130

A Bandgap Reference (BGR) is a key analog circuit used to generate a stable reference voltage that remains nearly constant despite variations in temperature, supply voltage, and process. Because of its robustness, it is widely used as a core building block in LDOs, ADCs, DACs, buck converters, PMICs, and sensor interface circuits.

This project implements a Bandgap Reference IP in SkyWater 130nm (SKY130) CMOS technology. The design targets a 0.9 V reference output with low power consumption, good temperature stability, and reliable operation over a wide supply range. The complete flow includes circuit design, schematic implementation, pre-layout simulation in Ngspice, performance characterization, and layout design in Magic.

Overview

The purpose of this project is to design and analyze a low-power bandgap reference circuit using the SKY130 PDK. The circuit is characterized across temperature and supply variations to verify its suitability for analog and mixed-signal applications.

The work includes:

- Circuit architecture and implementation of the BGR
- Block diagram and transistor-level schematic
- Pre-layout simulation using Ngspice
- Reference voltage variation with temperature
- Reference voltage variation with supply voltage
- Temperature coefficient and voltage coefficient analysis
- Startup behavior analysis
- On/off current analysis using enable control
- Full-custom layout implementation using Magic

Key Features

- Implemented in SkyWater 130nm CMOS technology
- Generates a stable 0.9 V reference
- Designed for low-power operation
- Operates over a 2 V to 5 V supply range
- Characterized over wide temperature variation
- Includes startup circuit analysis
- Includes enable-based current consumption analysis
- Physical layout completed using Magic VLSI

Performance Summary

Parameter	Value
Reference Voltage, VREF	0.9 V
Supply Current (max)	0.1 mA
Output Voltage Noise	0.2 mVP-P (0.1 Hz to 10 Hz)
Output Current Capability	1 mA
Temperature Coefficient (max)	50 ppm/°C
Input Voltage Range	2 V – 5 V
Line Regulation	1 mV/V
Load Regulation	1 mV/mA

Tools and Technology

- **PDK:** SkyWater SKY130
- **Schematic / Simulation:** Ngspice
- **Layout:** Magic VLSI
- **Technology Node:** 130 nm CMOS

Simulation and Analysis

The following analyses are included in this project:

1. Performance parameters and circuit implementation of the Bandgap Reference IP
2. Block diagram and schematic of the BGR
3. Pre-layout simulation using Ngspice
4. VBGP vs Temperature from -40°C to 140°C at $R_L = 100\text{ M}\Omega$
5. VBGP vs VDD from 2 V to 4 V at $R_L = 100\text{ M}\Omega$
6. Temperature Coefficient vs Temperature from -40°C to 140°C
7. Voltage Coefficient vs VDD from 2 V to 4 V
8. Startup time analysis of VBGP
9. On/Off current analysis with respect to enable signal
10. Layout implementation of the BGR using Magic

Why This Design Matters

Bandgap references are essential in modern integrated circuits because many analog and mixed-signal blocks require an accurate and temperature-insensitive voltage reference. A well-designed BGR improves the reliability and precision of larger systems such as voltage regulators, data converters, and power management circuits.

This project demonstrates a complete custom analog design flow in open-source SKY130 technology, making it useful for learning analog IC design, simulation methodology, and layout implementation in a practical environment.

Applications

This BGR design can be used as a building block in:

- Low Dropout Regulators (**LDOs**)
- Analog-to-Digital Converters (**ADCs**)
- Digital-to-Analog Converters (**DACs**)
- Buck / DC-DC converters
- Power Management ICs (**PMICs**)
- Sensor and mixed-signal interface circuits

Layout

The physical layout of the Bandgap Reference has been implemented using Magic in the SKY130 PDK. The layout is intended to validate that the circuit can be realized in silicon while maintaining compact area and analog performance.

Conclusion

This project presents the design and analysis of a 0.9 V Bandgap Reference in SKY130 technology. The circuit shows stable operation across supply and temperature variations, low current consumption, and suitability for use in analog and mixed-signal systems. It also demonstrates a complete open-source analog IC design flow from schematic to layout.